

Linear Algebra

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Author

MUHO CHOI

Email: sihyeon8240@gmail.com

GitHub: <https://github.com/sihyeon8240>

Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/math>

Feel free to open issues or pull requests for corrections.

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Preface

This textbook uses Lean 4 to formalize selected mathematical statements and proofs.

- **Tools.** Lean is an interactive theorem prover; Mathlib is a library of formalized mathematics. This edition uses Lean 4.33.1 and Mathlib v4.33.1 (revision [0df444a360ea](#)).
- **Reading the proofs.** Each paired statement or proof step gives a mathematical explanation followed by its Lean code. The explanations can be read independently; code fragments are read in order within the complete declaration.
- **Verification.** Lean checks formal proofs under their stated assumptions. Verification covers only statements linked to checked declarations. Agreement between the explanations, displayed code, and checked proofs is reviewed separately.
- **Prerequisites.** Starting assumptions and accepted external results are specific to this textbook.

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12.3 Extreme Points and Supporting Hyperplanes

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Bibliography

- [1] Serge Lang, *Linear Algebra*, Springer Nature.